Decent Coaching Classes

VIII SB

Mind Your Math VIII

ANSWER KEY **Marks: 30**

Q.1 (A) **(3)**

(1) (C) orthocentre (2) (B) x × y (3) (D) (x − 2)(x − 3)

Q.1 (B) **(2)**

(1) The centroid divides each median in the ratio 2 : 1 from the vertex. l(AG) = 2/3 × l(AD) = 2/3 × 12 = **8 cm**.

(2) $y \propto x$, so $y = kx$, so $20 = k \times 4$, so $\mathbf{k = 5}$.

Q.2 (any four, 2 marks each) **(8)**

(1) $x^2 + 9x + 20 = x^2 + 4x + 5x + 20 = x(x + 4) + 5(x + 4) = \mathbf{(x + 4)(x + 5)}$

(2) $x^3 - 27 = x^3 - 3^3$, and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$, so $\mathbf{(x - 3)(x^2 + 3x + 9)}$

(3) $A \propto r^2$ (1 mark); equation of variation: $\mathbf{A = kr^2}$, where k is the constant of variation (1 mark).

(4) Steps (any correct construction): (i) Draw an obtuse-angled Δ LMN. (ii) Find the mid-point of each side (using ruler/compass). (iii) Join each vertex to the mid-point of its opposite side to get the three medians. (iv) The medians meet at one point inside the triangle; name it G. (Triangle 1 mark, medians + centroid 1 mark.)

(5) $a \propto 1/b$, so $a \times b = k$. $k = 10 \times 6 = 60$. When $b = 4$: $a \times 4 = 60$, so $\mathbf{a = 15}$.

(6) $(x^2 - 9)/(x + 3) = (x - 3)(x + 3)/(x + 3) = \mathbf{x - 3}$

Q.3 (any three, 3 marks each) **(9)**

(1) $2x^2 - 7x - 15$: product of first and last coefficient = 2 × (−15) = −30; factors with sum −7 are −10 and +3.

$= 2x^2 - 10x + 3x - 15 = 2x(x - 5) + 3(x - 5) = \mathbf{(x - 5)(2x + 3)}$

(2) $54p^3 + 250q^3 = 2(27p^3 + 125q^3) = 2[(3p)^3 + (5q)^3]$

$= \mathbf{2(3p + 5q)(9p^2 - 15pq + 25q^2)}$

(3) Speed (s) and time (t) are in inverse variation, so $s \times t = k$. $k = 45 \times 4 = 180$. For $s = 60$: $60 \times t = 180$, so $\mathbf{t = 3\ hours}$.

(4) Let cost = x and weight = y. $x \propto y$, so $x = ky$. $225 = k \times 5$, so $k = 45$. Equation: $x = 45y$. For $y = 12$: $x = 45 \times 12 = \mathbf{₹\ 540}$.

(5) The centroid divides each median in the ratio 2 : 1. BG = 2 × GQ, so 10 = 2 × GQ, so **l(GQ) = 5 cm**. CG = 2 × GR = 2 × 3.5 = 7 cm. l(CR) = CG + GR = 7 + 3.5 = **10.5 cm**.

Q.4 (any two, 4 marks each) **(8)**

(1) $x^2 - 9 = (x - 3)(x + 3)$; $x^2 + 5x + 6 = (x + 2)(x + 3)$; $x^2 + 2x = x(x + 2)$; $x^2 - 3x = x(x - 3)$. (2 marks)

Expression = $[(x - 3)(x + 3)] / [(x + 2)(x + 3)] \times [x(x + 2)] / [x(x - 3)]$

$= \mathbf{1}$ (cancelling the common factors) (2 marks)

(2) (a) $6x^2 + 7x - 3$: $6 \times (-3) = -18$; factors with sum 7 are 9 and −2.

$6x^2 + 9x - 2x - 3 = 3x(2x + 3) - 1(2x + 3) = \mathbf{(2x + 3)(3x - 1)}$ (2 marks)

(b) $125a^3 - 8b^3 = (5a)^3 - (2b)^3 = \mathbf{(5a - 2b)(25a^2 + 10ab + 4b^2)}$ (2 marks)

(3) Steps:

(i) Draw seg YZ = 6 cm. With Y as centre and radius 5 cm, and Z as centre and radius 7 cm, draw arcs intersecting at X. Join XY and XZ. (Δ XYZ, 1 mark)

(ii) Using a set-square, draw perpendiculars from each vertex to the opposite side: XP $\perp$ YZ, YQ $\perp$ XZ and ZR $\perp$ XY. (2 marks)

(iii) The three altitudes are concurrent. Name the point of concurrence O; it is the orthocentre. (1 mark)

Note: $7^2 = 49 < 5^2 + 6^2 = 61$, so the triangle is acute-angled and O lies inside it.